

Zephyr RTOS Porting Comparison

Technical Hardware & Memory Mapping Analysis

Reference: PIC32CM GC00 → Target: PIC32CM5164LS0048

REFERENCE SOC

PIC32CM GC00 (100-pin)

TARGET SOC

PIC32CM5164LS0048

TARGET BOARD

PIC32CM LS Curiosity

EXECUTIVE PORTING SUMMARY HIGHLIGHTS

Flash & Core Sizing

512 KB Flash mapped at `0x00000000`. Core Cortex-M23 execution environment is 100% compatible.

SRAM & System Clock

64 KB SRAM (down from 128 KB). System max clock reduced from **72 MHz to 48 MHz**.

Security & TrustZone

LS00 introduces **ARM TrustZone-M**. Requires `CPU_HAS_ARM_TRUSTZONE_M` Kconfig and SAU nodes.

Peripheral & Driver Reuse

Direct reuse for SERCOM (UART/SPI/I2C), GPIO, Timers, WDT, ADC/DAC. CAN-FD is removed.

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Parameter	PIC32CM GC00 (Reference in Zephyr)	PIC32CM5164LS0048 (Your Target)	Change Required?
PORT Groups	PORTA, PORTB, PORTC	PORTA, PORTB only	⚠ Remove PORTC node in DTS
XOSC (Crystal)	Up to 32 MHz external	Up to 32 MHz external	✓ Same
DFLL48M	Yes	Yes	✓ Same
DPLL	Yes (PLL0, PLL1)	Yes (FDPLL96M)	⚠ LS00 uses different PLL naming
Power Levels	PL0/PL2 required for peripherals	PL0/PL2 required for peripherals	✓ Same errata applies
Sleep Modes	Idle, Standby, Hibernate, Backup	Idle, Standby, Hibernate, Backup	✓ Same

PART 2 — Memory Map: Exact Hex Address Comparison

Core Memory Regions

Region	GC00 Address / Size	LS0048 Address / Size	Action
Flash Start	0x00000000	0x00000000	✓ Same
Boot Flash	0x00000000 , 16 KB	0x00000000 , 16 KB (configurable BOOTPROT)	✓ Same
Application Flash	0x00004000 , 496 KB	0x00004000 , 496 KB	✓ Same
Flash End	0x0007FFFF	0x0007FFFF	✓ Same
SRAM Start	0x20000000	0x20000000	✓ Same
SRAM Size	0x20000 (128 KB)	0x10000 (64 KB)	⚠ MUST change reg
SRAM End	0x2001FFFF	0x2000FFFF	⚠ Update in DTS
TrustRAM	0x47000000 (512 B)	0x47000000 (512 B)	✓ Same
Data Flash (NVM)	0x00400000	0x00400000	✓ Same

System Peripheral Base Addresses

Peripheral	GC00 Base Address	LS0048 Base Address	Action
PAC	0x40000000	0x40000000	✓ Same
FREQM	0x40000C00	0x40000C00	✓ Same
WDT	0x40001000	0x40001000	✓ Same
RTC	0x40001400	0x40001400	✓ Same
EIC	0x40001800	0x40001800	✓ Same
MCLK	0x40000800	0x40000800	✓ Same
OSCCTRL	0x40001000	0x40001000	✓ Same
OSC32KCTRL	0x40001400	0x40001400	✓ Same
SUPC	0x40001800	0x40001800	✓ Same
GCLK	0x40001C00	0x40001C00	✓ Same
PM	0x40000400	0x40000400	✓ Same

APB Bridge A Peripherals

Peripheral	GC00 Base Address	LS0048 Base Address	Action
PORT	0x41000000	0x41000000	✓ Same
DSU	0x41002000	0x41002000	✓ Same
NVMCTRL	0x41004000	0x41004000	✓ Same

APB Bridge B Peripherals: SERCOM Instances

Peripheral	GC00 Base Address	LS0048 Base Address	Action
SERCOM0	0x42000000	0x42000000	✓ Same
SERCOM1	0x42000400	0x42000400	✓ Same
SERCOM2	0x42000800	0x42000800	✓ Same
SERCOM3	0x42000C00	0x42000C00	✓ Same
SERCOM4	0x42001000	0x42001000	✓ Same
SERCOM5	0x42001400	0x42001400	✓ Same

APB Bridge B Peripherals: Timers

Peripheral	GC00 Base Address	LS0048 Base Address	Action
TCC0	0x42001800	0x42001800	✓ Same
TCC1	0x42001C00	0x42001C00	✓ Same
TCC2	0x42002000	0x42002000	✓ Same
TC0	0x42002400	0x42002400	✓ Same
TC1	0x42002800	0x42002800	✓ Same
TC2	0x42002C00	0x42002C00	✓ Same
TC3	0x42003000	0x42003000	✓ Same
TC4	0x42003400	0x42003400	✓ Same

APB Bridge B: Analog & Other

Peripheral	GC00 Base Address	LS0048 Base Address	Action
ADC	0x42003800	0x42003800	✓ Same
AC	0x42003C00	0x42003C00	✓ Same
DAC	0x42004000	0x42004000	✓ Same
PTC	0x42004400	0x42004400	✓ Same
DMAC	0x4101C000	0x4101C000	✓ Same
USB	0x41000000	0x41000000	✓ Same
CAN0	0x42001C00	Not Present	⚠ Remove CAN0 node
CAN1	0x42002000	Not Present	⚠ Remove CAN1 node
I2S	Not Present	0x42004800	⚠ Add I2S node for LS00
OPAMP	Not Present	0x42004C00	⚠ Add OPAMP node for LS00

★ IMPORTANT

The base addresses above must be verified against the actual LS00 datasheet (DS60001615) pages covering "Product Mapping" / "Peripheral Summary", as some minor offsets may differ on sub-bridge assignments compared to GC00 (DS60001916).

PART 3 — NVIC Interrupt Number Comparison

Interrupt Source	GC00 IRQ#	LS0048 IRQ#	Notes
EIC EXTINT0	IRQ 12	IRQ 12	✓ Same
DMAC CH0	IRQ 31	IRQ 31	✓ Same
USB	IRQ 34	IRQ 34	✓ Same
SERCOM0 (0-3)	IRQ 65–68	IRQ 65–68	✓ Same
SERCOM1 (0-3)	IRQ 69–72	IRQ 69–72	✓ Same
SERCOM2 (0-3)	IRQ 73–76	IRQ 73–76	✓ Same
SERCOM3 (0-3)	IRQ 77–80	IRQ 77–80	✓ Same
SERCOM4 (0-3)	IRQ 81–84	IRQ 81–84	✓ Same
SERCOM5 (0-3)	IRQ 85–88	IRQ 85–88	✓ Same
TCC0 (0-3)	IRQ 89–92	IRQ 89–92	✓ Same
TCC1 (0-3)	IRQ 93–96	IRQ 93–96	✓ Same
TCC2 (0-3)	IRQ 97–100	IRQ 97–100	✓ Same
TC0	IRQ 101	IRQ 101	✓ Same
TC1	IRQ 102	IRQ 102	✓ Same
ADC (0+1)	IRQ 107-108	IRQ 107-108	✓ Same
DAC	IRQ 110	IRQ 110	✓ Same
CAN0	IRQ 119	Not Present	⚠ No CAN in LS00
CAN1	IRQ 120	Not Present	⚠ No CAN in LS00

⚠ WARNING

The IRQ numbers above are derived from typical Cortex-M23 PIC32CM family patterns. **Always verify** exact IRQ numbers from the LS0048 datasheet Interrupt Controller table before writing the DTS `interrupts` property.

PART 4 — Zephyr Directory Structure Comparison

GC00 Existing Layout (your template)

```
zephyr/
├── soc/microchip/pic32c/pic32cm_sg_gc/
│   ├── CMakeLists.txt
│   ├── Kconfig
│   ├── Kconfig.defconfig
│   ├── Kconfig.soc
│   ├── soc.yml
│   ├── common/
│   │   ├── pic32cm_sg_gc_100.dtsi    ← 100-pin pinout
│   │   ├── pic32cm_sg_gc_64.dtsi    ← 64-pin pinout
│   │   ├── pic32cm_sg_gc_48.dtsi    ← 48-pin pinout
│   │   └── pic32cm_5112_sg_gc.dtsi  ← memory sizing
│   ├── pic32cm_gc00/
│   │   └── pic32cm5112gc00100.dtsi  ← chip-specific dtsi
│   └── pic32cm_sg00/
│       └── pic32cm5112sg00100.dtsi
├── dts/arm/microchip/pic32c/pic32cm_sg_gc/ ← DTS include path
└── boards/microchip/pic32c/pic32cm_gc00_cpro/
    ├── board.yml
    ├── pic32cm_gc00_cpro.dts
    └── pic32cm_gc00_cpro_defconfig
```

LS00 New Layout (what you will create)

```
zephyr/
├── soc/microchip/pic32c/pic32cm_ls/    ← NEW directory
│   ├── CMakeLists.txt                ← copy + change series name
│   ├── Kconfig                      ← change SOC_FAMILY name
│   ├── Kconfig.defconfig             ← change SOC_FAMILY name
│   ├── Kconfig.soc                  ← change SOC_FAMILY name
│   ├── soc.yml                      ← list ls00 SoC variants
│   ├── common/
│   │   ├── pic32cm_ls_48.dtsi        ← 48-pin peripherals layout
│   │   └── pic32cm_5164_ls.dtsi      ← 512KB Flash, 64KB SRAM
│   ├── pic32cm_ls00/
│   │   └── pic32cm5164ls00048.dtsi  ← your chip-specific dtsi
├── boards/microchip/pic32c/pic32cm_ls_curiosity/ ← NEW board dir
│   ├── board.yml
│   ├── pic32cm_ls_curiosity.dts
│   └── pic32cm_ls_curiosity_defconfig
```

PART 5 — Kconfig Changes (Line by Line)

File: `Kconfig.soc`

GC00	LS00	Notes
<code>SOC_FAMILY_MICROCHIP_PIC32CM_SG_GC</code>	<code>SOC_FAMILY_MICROCHIP_PIC32CM_LS</code>	Rename family
<code>"microchip_pic32cm_sg_gc"</code>	<code>"microchip_pic32cm_ls"</code>	Rename string

File: **Kconfig**

GC00 Setting	LS00 Setting	Notes
SOC_FAMILY_MICROCHIP_PIC32CM_SG_GC	SOC_FAMILY_MICROCHIP_PIC32CM_LS	Rename
select CPU_CORTEX_M23	select CPU_CORTEX_M23	✓ Keep
select CPU_HAS_ARM_MPU	select CPU_HAS_ARM_MPU	✓ Keep
No TrustZone	select CPU_HAS_ARM_TRUSTZONE_M	⚠ ADD TrustZone config
select CLOCK_CONTROL	select CLOCK_CONTROL	✓ Keep
No I2S	select I2S	⚠ Optionally ADD

File: **soc.yaml**

GC00 Entry	LS00 Entry
family: microchip_pic32cm_sg_gc	family: microchip_pic32cm_ls
series: pic32cm_gc00	series: pic32cm_ls00
socs: pic32cm5112gc00048/064/100	socs: pic32cm5164ls00048

PART 6 — DeviceTree **.dtsi** Node Changes

Memory Nodes

DTS Property	GC00 Value	LS00 Value	Action
sram0 reg	<0x20000000 DT_SIZE_K(128)>	<0x20000000 DT_SIZE_K(64)>	⚠ Change SRAM
flash0 reg	<0x00000000 DT_SIZE_K(512)>	<0x00000000 DT_SIZE_K(512)>	✓ Same
boot_flash reg	<0x0 DT_SIZE_K(16)>	<0x0 DT_SIZE_K(16)>	✓ Same
program_flash reg	<0x0 DT_SIZE_K(496)>	<0x0 DT_SIZE_K(496)>	✓ Same

CPU / Clock Node

DTS Property	GC00 Value	LS00 Value	Action
clock-frequency	<72000000>	<48000000>	⚠ Must change
compatible (clock)	"microchip,pic32cm-sg-gc-clock"	"microchip,pic32cm-ls-clock"	⚠ New clock binding
DPLL source	dp110/dp111	fdp1196m	⚠ Different PLL naming

PORT (GPIO) Nodes

GC00 Nodes	LS00 Nodes	Action
porta	porta	✓ Keep
portb	portb	✓ Keep
portc	Remove	⚠ 48-pin has no PORTC

Peripheral Nodes to REMOVE for LS00

Node	Reason
can0	LS00 has no CAN
can1	LS00 has no CAN
portc	48-pin package has no PORTC

Peripheral Nodes to ADD for LS00

Node	Reason
i2s: i2s@42004800	LS00 has I2S peripheral
opamp: opamp@42004C00	LS00 has OPAMP
TrustZone SAU/IDAU nodes	TrustZone memory attribution

SERCOM Nodes (All Same - Copy Directly)

```
sercom0 @ 0x42000000 → compatible = "microchip,pic32cm-sercom" ✓ Same
sercom1 @ 0x42000400 → compatible = "microchip,pic32cm-sercom" ✓ Same
sercom2 @ 0x42000800 → compatible = "microchip,pic32cm-sercom" ✓ Same
sercom3 @ 0x42000C00 → compatible = "microchip,pic32cm-sercom" ✓ Same
sercom4 @ 0x42001000 → compatible = "microchip,pic32cm-sercom" ✓ Same
sercom5 @ 0x42001400 → compatible = "microchip,pic32cm-sercom" ✓ Same
```

PART 7 — Board DTS Comparison (board .dts file)

Parameter	GC00 Curiosity Pro	LS00 Curiosity (New)	Notes
model	"PIC32CM GC00 Curiosity Pro"	"PIC32CM LS00 Curiosity"	Update string
compatible	"microchip,pic32cm_gc00_cpro"	"microchip,pic32cm_ls_curiosity"	New compat string
socs (board.yml)	pic32cm5112gc00100	pic32cm5164ls00048	Update to LS00 part
clock-frequency	72000000	48000000	Reduce clock
LED0 GPIO pin	&porta 20	Verify with your schematic	Check board schematic
BUTTON0 GPIO	&portb 7	Verify with your schematic	Check board schematic

PART 8 — Driver Reuse Strategy

Driver / Subsystem	GC00 Uses	LS00 Can Reuse?	Notes
GPIO	<code>microchip-pic32c-gpio.c</code>	✓ Yes - direct reuse	Same PORT IP
UART (SERCOM)	<code>uart_sam0.c</code>	✓ Yes - direct reuse	Same SERCOM IP
SPI (SERCOM)	<code>spi_sam0.c</code>	✓ Yes - direct reuse	Same SERCOM IP
I2C (SERCOM)	<code>i2c_sam0.c</code>	✓ Yes - direct reuse	Same SERCOM IP
Clock Control	<code>clock_control_pic32cm_sg_gc.c</code>	⚠ Need new driver	Different clock bindings
Pin Control	<code>pinctrl-pic32c.c</code>	✓ Yes - direct reuse	Same PORT pinmux IP
Flash (NVMCTRL)	<code>flash_sam0.c</code>	✓ Yes - direct reuse	Same NVM controller
WDT	<code>wdt_sam0.c</code>	✓ Yes - direct reuse	Same WDT IP
RTC	<code>rtc_sam0.c</code>	✓ Yes - direct reuse	Same RTC IP
USB	<code>usb_dc_sam0.c</code>	✓ Yes - direct reuse	Same USB IP
ADC	<code>adc_sam0.c</code>	✓ Yes - direct reuse	Same ADC IP
DAC	<code>dac_sam0.c</code>	✓ Yes - direct reuse	Same IP
DMA	<code>dmac_sam0.c</code>	✓ Yes - direct reuse	Same IP
CAN	<code>can_mcan.c</code>	✗ Not needed	LS00 has no CAN
I2S	Not in GC00	⚠ May need new driver	LS00 has I2S
TrustZone	Not in GC00	⚠ Need TZ config	New security setup

PART 9 — Clock Architecture Comparison

GC00 Clock Tree (complex DPLL-based 72 MHz)

XOSC (12 MHz) → DPLL0 (×12 = 144 MHz) → /2 = 72 MHz → CPU
→ DFLL48M (×4 = 48 MHz) → USB
XOSC32K → GCLK generators → Peripheral clocks

LS00 Clock Tree (FDPLL96M-based 48 MHz)

XOSC (depends on board crystal) → FDPLL96M (up to 96 MHz) → /2 = 48 MHz → CPU
DFLL48M → USB
OSCULP32K / XOSC32K → RTC, WDT clocks
GCLK generators → SERCOM, TC, TCC, ADC clocks (via MCLK enable bits)

Key Difference: The GC00 uses DPLL0/DPLL1 terminology while the LS00 uses FDPLL96M. The Zephyr clock control driver needs a new binding for LS00's clock system.

PART 10 — Summary: What to Copy vs. What to Modify

Zephyr File	Action	Detail
<code>soc/microchip/pic32c/pic32cm_sg_gc/Kconfig</code>	Copy + Modify	Rename SOC family, add TrustZone selects
<code>soc/microchip/pic32c/pic32cm_sg_gc/Kconfig.soc</code>	Copy + Modify	Rename family string
<code>soc/microchip/pic32c/pic32cm_sg_gc/CMakeLists.txt</code>	Copy + Modify	Change directory references
<code>soc/microchip/pic32c/pic32cm_sg_gc/soc.yml</code>	Copy + Modify	List LS00 SoC variants
<code>common/pic32cm_sg_gc_100.dtsi</code>	Create new	<code>pic32cm_ls_48.dtsi</code> with 48-pin peripherals
<code>common/pic32cm_5112_sg_gc.dtsi</code>	Copy + Modify	Change SRAM to 64KB, clock to 48MHz
<code>pic32cm_gc00/pic32cm5112gc00100.dtsi</code>	Create new	<code>pic32cm5164ls00048.dtsi</code>
<code>boards/.../pic32cm_gc00_cpro.dts</code>	Create new	<code>pic32cm_ls_curiosity.dts</code>
<code>boards/.../board.yml</code>	Create new	Reference new SoC name
<code>boards/.../*_defconfig</code>	Create new	48 MHz system clock setting
Clock driver	Create new	New binding for FDPLL96M + LS00 clock scheme
All other drivers	Reuse	Direct compatibility with LS00 SERCOM/PORT IP